

Sarah A. Kusumastuti, Ph.D

*Cognition, Data, and Education (CoDE) Section
Department of Learning, Data analytics and Technology (LDT)
Faculty of Behavioural, Management and Social Sciences (BMS)
University of Twente*

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EDUCATION

University of Southern California, Los Angeles, CA

PhD in Quantitative Psychology

May 2021

Advisor: Dr. Richard S John

Dissertation: *"The Cost of Not Identifying Objectives in Multiattribute Decision Modeling"*

Master of Arts (MA) in Quantitative Psychology

August 2016

Advisor: Dr. Richard S John

Thesis: *"Trade-offs among attributes of Authentication"*

Purdue University, West Lafayette, IN

Bachelor of Science (BSc) in Mathematical Statistics

May 2014

Minors in Psychology and Philosophy

RESEARCH INTERESTS

Judgment and decision making, human factors, human-in-the-loop simulations, experiment design, decision analysis, multiattribute value/utility analysis, computational social science, transportation research, behavioural game theory, behavioural cybersecurity, risk perception

EMPLOYMENT

University of Twente - Enschede, The Netherlands

Assistant Professor

2025 - present (permanent since 2026)

- Cognition, Data analytics, and Education (CoDE) Section, Department of Learning, Data Analytics and Technology (LDT), Faculty of Behavioural Management and Social Sciences (BMS)
- SSH Sector Plan member for The Human factor In New Technologies (HINT)

University of Twente - Enschede, The Netherlands

Postdoctoral Researcher

2023 - 2025

- Cognition, Data analytics, and Education (CoDE) Section, Department of Learning, Data Analytics and Technology (LDT), Faculty of Behavioural Management and Social Sciences (BMS) in collaboration with **ProRail B.V.** for **Europe's Rail Joint Undertaking (ERJU)**

University of Groningen - Groningen, The Netherlands

Postdoctoral Researcher

2022 - 2023

- Department of Innovation Management & Strategy, Faculty of Economics and Business

RESEARCH EXPERIENCE

DEVCOM Army Research Lab (ARL) West - <i>Journeyman Fellow</i>	2020-2021
USC Institute of Creative Technologies (USC-ICT) - <i>Summer Research Intern</i>	2020
Center for Risk Analysis of Terrorism Events (CREATE) - <i>Research Assistant</i>	2014 - 2021
Traveloka - <i>Design Research Intern</i>	2018
USC Information Sciences Institute (USC-ISI) - <i>Research Assistant</i>	2015-2016
Sandia Research Corporation - <i>Research Assistant</i>	2015-2016

PEER REVIEWED PUBLICATIONS

JOURNAL ARTICLES

1. Borsci, S., Kusumastuti, S.*, Lo, J., & Kolkman, T. (2026). Acceptability of autonomous trains across Europe: A multinational survey of preferences, support, and intention to use. Manuscript under revision. <https://doi.org/10.2139/ssrn.6966675>
2. Kusumastuti, S. A.*, Kolkman, T. H., Lo, J. C., & Borsci, S. (2025). Charting the landscape of rail human factors and automation: A systematic scoping review. *Transportation Research Interdisciplinary Perspectives*, 30, 101350. <https://doi.org/10.1016/j.trip.2025.101350>
3. Kusumastuti, S. A.*, & John, R. S. (2024). A simulation approach to investigate factors influencing the cost of omitted objectives in multiattribute models. *Journal of Multi-Criteria Decision Analysis*, 31(1-2), e1826. <https://doi.org/10.1002/mcda.1826>
4. Kusumastuti, S.A.*, Pollard, K.A., Oiknine, A.H., Dalangin, B., Raber, T.R., & Files, B.T. (2022). Practice improves performance of a 2D uncertainty integration task within and across visualizations. *IEEE Transactions on Visualization and Computer Graphics*. <https://doi.org/10.1109/TVCG.2022.3173889>
5. Kusumastuti, S.A.*, Blythe, J., Rosoff, H. & John, R.S. (2020). Behavioral determinants of target shifting and deterrence in an analog cyber-attack game. *Risk Analysis*. <https://doi.org/10.1111/risa.13402>
6. Kusumastuti, S. *, Rosoff, H., & John, R.S. (2019). Characterizing conflicting user values for cyber authentication using a virtual public values forum. *Decision Analysis*. <https://doi.org/10.1287/deca.2018.0383>

BOOK CHAPTERS

1. Kusumastuti, S.A.*, John, R.S. (2024). Consequences of Missing Objectives in Applications of Multiattribute Utility Analysis. In: Federspiel, F.M., Montibeller, G., Seifert, M. (eds) *Behavioral Decision Analysis. International Series in Operations Research & Management Science, vol 350*. Springer, Cham. https://doi.org/10.1007/978-3-031-44424-1_9

CONFERENCE PROCEEDINGS

1. Kusumastuti, S.*, Kolkman, T., Lo, J., Borsci, S. (2026). Development of Human Factors toolkit to inform behavioural and cognitive research in the railway domain. In: Dimitrios Ziakkas, Riccardo Patriarca, Amit Sharma and Steven Mallam (eds) *Advances in Human Factors of Transportation. AHFE (2026) International Conference*. AHFE Open Access, vol 224. AHFE International, USA.
<https://doi.org/10.54941/ahfe1007869>
2. Kolkman, T., Kusumastuti, S. *, Borsci, S., Lo, J. (2026). Operational Transitions to Automation: A Scoping review with implications for future rail service. In: Dimitrios Ziakkas, Riccardo Patriarca, Amit Sharma and Steven Mallam (eds) *Advances in Human Factors of Transportation. AHFE (2026) International Conference*. AHFE Open Access, vol 224. AHFE International, USA.
<https://doi.org/10.54941/ahfe1007872>
3. Kusumastuti, S. A. *, Lo, J. C., Kolkman, T., & Borsci, S. (2026) A multi-actor human-in-the-loop simulation methodology to examine human-automation interaction in rail: Lessons learned from a simulation of Automatic Train Operations (ATO) Grade of Automation (GoA) level 2 systems. In D. Golightly, R. Charles & F. Dixon (eds) *Contemporary Ergonomics and Human Factors 2026*. CIEHF, Nottingham, UK
<https://publications.ergonomics.org.uk/publications/a-multi-actor-human-in-the-loop-simulation-methodology-to-examine-human-automation-interaction-in-railways>

GRANTS AND AWARDS

- *Alliance VU-UT Seed Grant - Smart Societies 2026*
Role: Co-PI with Russel Chan (University of Twente) and Yannick Hill (Vrije Universiteit Amsterdam)
Amount: €35,000
Title: Performance Under Pressure (PUP): A Pilot Application of Personalised Stress Profile in First Responders During Emergency Training
- *First Place - Chevron Pacific Indonesia Scholastic Awards (2010-2014)*
Amount: \$189,000 for 4.5 years to pursue bachelor's degree, covering college tuition and living expenses
- *Summer Research Grant - Department of Psychology, University of Southern California* (Summer 2016 & Summer 2017)
Amount: \$6000

HONOURS AND ACTIVITIES

- *Associate Fellow* - Department of Innovation & Organization (I&O), Faculty of Economics and Business Research Institute (FEBRI), University of Groningen (2022)
- *Journeyman Fellowship* - CCDC Army Research Laboratory (ARL) (2020-2021)
- *Semester Honors* - College of Science, Purdue University (2011-2014)
- *Indonesian National Science Olympiad (OSN) in Astronomy*
 - Bronze medalist (2008) - Makassar, South Sulawesi
 - Silver medalist (2009) - DKI Jakarta

CONFERENCE CONTRIBUTIONS

Papers Presented

1. Kusumastuti, S., Cui, T., Tambe, A., & John, R.S. (2015, March). *A behavioral game modeling cyber attackers, defenders, and users*. AAAI Spring Symposium 2015 on Applied Computational Game Theory.

Oral Presentations

1. Kusumastuti, S. A., Kolkman, T. H., Lo, J. C., & Borsci, S. (2025, February 18-19). *Development of human factors toolkit to inform behavioural research in the railway domain* [Conference session]. 6th German Conference on Rail Human Factors, Berlin, Germany.
<https://rhf2025.bbi-berlin.com/program.html>
2. Kusumastuti, S., Lo, J. & Borsci, S. (2023, November 23). *The future of rail operators in the advent of Automated Train Operations (ATO)* [Conference session]. Human Factors NL Jaarcongres 2023, Enschede, Netherlands.
3. Kusumastuti, S. (2022, June 22-24). *The cost of missing objectives in multiattribute decision modeling* [Conference session]. 2022 Advances in Decision Analysis Conference, Rosslyn, Virginia, USA.
4. Kusumastuti, S. (2019, December 8-12). *Measuring the consequences of incomplete objective sets using a Monte Carlo approach in selected applications of multiattribute models and simulated datasets* [Conference session]. Society for Risk Analysis Annual Meeting 2019, Arlington, Virginia, USA.
5. Kusumastuti, S. (2019, October 20-23). *A simulation approach to compare decision quality of multiattribute models with incomplete attribute sets* [Conference session]. INFORMS Annual Meeting 2019, Seattle, Washington, USA.
6. Kusumastuti, S., & John, R.S. (2018, December 2-6). *An Empirical Behavioral Study of Deterrence: An Analog Cyber Attack Simulation Game* [Conference session]. Society for Risk Analysis Annual Meeting 2018, New Orleans, Louisiana, USA.
7. John, R. S., Kusumastuti, S., Rosoff, H., & Blythe, J. (2018, November 4-7). *Concentrated vs. distributed defense as a deterrent to cyber attackers: A behavioral analog simulation game* [Conference session]. INFORMS Annual Meeting 2018, Phoenix, Arizona, USA.
8. Kusumastuti, S., Rosoff, H., & John, R.S. (2016, December 11-15). *An interactive real-time behavioral game for cyber security* [Conference session]. Society for Risk Analysis Annual Meeting 2016, San Diego, California, USA.
9. Rosoff, H., Blythe, J., Kusumastuti, S., & John, R.S. (2016, December 11-15). *Behavioral experimentation of cyber attacker deterrence with DETER testbed* [Conference session]. Society for Risk Analysis Annual Meeting 2016, San Diego, California, USA.
10. Kusumastuti, S., Rosoff, H., & John, R.S. (2015, December 6-10). *Tradeoff value assessment for features of password as a security measure and method of authentication* [Conference session]. Society for Risk Analysis Annual Meeting 2015, Arlington, Virginia, USA.

Poster Presentations

1. Kusumastuti, S. A., Kolkman, T. H., Lo, J. C., & Borsci, S. (2025, September 18-19). *Introducing a human factors toolkit to inform research utilising Human-in-the-Loop (HITL) simulation in the railway domain*. [Poster session]. International Human Factors Rail Conference 2025, London, United Kingdom.
2. Kusumastuti, S., Blythe, J., Rosoff, H., & John, R.S. (2019, August 2-5). *Deterrence as target shifting in layered cyber defense: A behavioral analog game* [Poster session]. The Second Conference on Risk Analysis, Decision Analysis and Security. Buffalo/Niagara Falls, New York, USA
3. Kusumastuti, S., Rosoff, H., & John, R.S. (2015, November 20-23). *The effects of risk-related perception on password preferences and behavior* [Poster session]. Society of Judgment and Decision Making, Chicago, Illinois, USA.
4. Kusumastuti, S., Guney, S., & John, R.S. (2014, December). *A behavioral game modeling cyber attackers, defenders, and users* [Poster session]. The 4th Annual Review of ARO Game Theory MURI, University of Southern California, Los Angeles, California, USA.

INVITED TALKS

- 2026 - “The Cost of Not Identifying Objectives.” Invited speaker at Delft University of Technology, Department of Technology, Policy, and Management (TPM)
- 2015 - “Designing factorial experiments.” Lecture given at the Graduate Consultation and Computer Center (GC3), University of Southern California, Los Angeles, CA.

STUDENT SUPERVISION

University of Twente

MSc. students: Jielong de Ridder (2027), Nynke Kruwel (2026), Bram Brinkman (2026) (co-supervision with Simone Borsci), Leonie Döhrmann & Nikola Markiewicz (2025) (co-supervision with Simone Borsci)
BSc. students: Nynke Herik (2026), Sienna Petersma & Shalina Dirr (2026) (co-supervision with Funda Yildirim), Sytse Vos (2025) (co-supervision with Simone Broschi)

TEACHING EXPERIENCE

University of Twente

Data Analysis

Lecturer and Tutorial lead/assistant

- Data Analysis I & II
- Data Collection and Test Construction
- Research Methodology and Descriptive Statistics

Lecturer

- Smart Mobility - Human-AI teaming in mobility

University of Southern California (Graduate Teaching Assistant)

- PSYC339: Origins of the Mind
- PSYC316: Non Experimental Research Methods
- PSYC274: Statistics

Purdue University (Undergraduate Teaching Assistant)

- MA137: Mathematics for Elementary Teachers I
- MA138: Mathematics for Elementary Teachers II
- STAT417: Statistical Theory

SERVICE

Peer Reviewer

- Journal: *Risk Analysis*, *Decision Analysis*, *IEEE Access*, *Frontiers in Psychology*, *Frontiers in Future Transportation*, *Frontiers in Bionformatics* and *Terrorism and Political Violence*
- Grant: Nazarbayev University Research Review - 2026 CRP Cycle

Outreach

- *Kijk in de toekomst bij Universiteit Twente*: Presenting simulation facilities at the University of Twente BMS Lab to the public as part of Enschede700 initiative (2025)
<https://www.utoday.nl/video/76099/bms-lab-geeft-burgers-kijkje-in-de-keuken>

SKILLS

Programming: R, Python, SAS, MATLAB, Java, Javascript, parallel and high performance computing (HPC)

Languages: English (full working proficiency), Indonesian (native), Dutch (B1), German (A1), Japanese (N4)

Multimedia: Adobe Photoshop, Adobe PremierePro, Adobe Illustrator, GIMP, Blender 3D